THE DRAKKAR ASCENT CHALLENGE

House Veridian Design Review Board — Structural Optimization of Booster Thrust Barrel

Cadre: Lockheed Martian | **Course:** Aircraft Structures (R5ME3203T) | **VJTI Mumbai**

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Mission Envelope & Applied Flight Loads (LF = 1.1)

- **Ascent Flight Condition:** Max dynamic pressure q = 24.5 kPa at V_inf = 200 m/s with severe 20 m/s vertical desert thermal gust.
- **Team Adopted Design Factor:** LF = 1.1 applied once directly to all primary flight forces and pressure before second-order analysis.
- **Axial Engine Compression:** F = 30.0 kN service --> 33.0 kN design thrust transmitted axially through the cylindrical shell.
- **Propellant Tank Ullage Pressure:** p = 0.30 MPa service --> 0.33 MPa design wall pressure (no beneficial pressure stabilization credited in buckling).
- **Aerodynamic Gust Side Loads:** Shear V = 1,465.1 N --> 1,611.6 N design; First-order Bending M_0 = 2,197.7 N.m --> 2,417.5 N.m design.
- **Beam-Column Amplified Moment:** P-delta moment magnification: M_design = 3,027.9 N.m (amplification factor = 1.252).

Theoretical Analysis Architecture (No FEA)

- **Exact Annular Geometry:** Exact midsurface radius r = (D - t)/2, area A = pi*(ro^2 - ri^2), and moment of inertia I = (pi/4)*(ro^4 - ri^4).
- **Second-Order Beam-Column Secant:** Exact pinned beam-column solution M_design = M_0 / cos[(L/2)*sqrt(P/EI)] eliminates heuristic P_eq approximations.
- **Co-Located Stress Envelope:** Evaluates signed axial compression, outer-fiber bending, hoop tension, and shear at identical circumferential positions (theta = 180 deg governs).
- **NASA SP-8007 Shell Buckling:** Empirical curvature knockdowns gx and gb as continuous functions of sqrt(r/t)/16 (gx = 0.651, gb = 0.717).
- **Combined Stability Interaction:** Tri-axial interaction criterion: (sigma_axial/sx_cr) + (sigma_bending/sb_cr) + (tau/tau_cr) <= 1.0.
- **Brazier Ovalization Diagnostic:** Classical elastic tube flattening capacity M_Brazier = 0.987*E*r*t^2/sqrt(1 - nu^2) evaluated at midspan.

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Parametric Material Trade Study (LF = 1.1)

Material	Class	Optimum OD	Wall (t)	Shell Mass	Status / Selection
Aluminium 7075-T6	Metal	119 mm	0.96 mm	1.501 kg	PRIMARY SELECTED WINNER
Magnesium AZ31B	Metal	169 mm	1.10 mm	1.540 kg	Heavier (+2.6%, yield cap)
Titanium Ti-6Al-4V	Metal	96 mm	0.84 mm	1.669 kg	Dense (+11.2%)
Steel 4130	Metal	154 mm	0.58 mm	3.292 kg	Heavy (+119.3%)
Carbon/Epoxy QI	Composite	101 mm	1.20 mm	0.903 kg	Theoretical min (unvalidated)
Aramid/Epoxy QI	Composite	169 mm	1.23 mm	1.361 kg	Laminate unvalidated
Glass/Epoxy QI	Composite	125 mm	1.60 mm	1.768 kg	Laminate unvalidated

Engineering Selection Rationale: Aluminium 7075-T6 offers high stiffness (71.7 GPa) and yield strength (503 MPa), beating Magnesium AZ31B by 2.6%. While Carbon/Epoxy QI is theoretically lighter, the brief lacks laminate ply schedules and delamination knockdowns required for certified aerospace production.

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Final Architecture: Aluminium 7075-T6 Thrust Barrel

• **Physical Dimensions:** Outside Diameter = 119 mm | Nominal Wall = 0.96 mm | Length = 1500 mm | Mean Radius r = 59.02 mm.

• **Component Mass:** Shell Mass = 1.500541 kg (~1.501 kg). Valid thin-walled cylinder: r/t = 61.5, L/r = 25.4.

• **Local Shell Buckling (GOVERNS):** Interaction Utilisation U = 0.9962 (Margin of Safety = +0.38%). Directly sizes nominal wall thickness.

• **Material Yielding Check:** Peak von Mises Stress = 393.9 MPa <= 503.0 MPa allowable. Utilisation U = 0.7830 (Margin of Safety = +27.7%).

• **Global Column Euler Buckling:** Critical Euler Thrust P_cr = 195.0 kN >> 33.0 kN design load. Utilisation U = 0.1692 (Margin of Safety = +491%).

• **Cross-Section Ovalization:** Max Bending Moment = 3,027.9 N.m < M_Brazier = 4,077.7 N.m. Utilisation U = 0.7425 (Margin of Safety = +34.7%).

• **Separate Transfer Load Case:** Carrying shear across length (M + V*L = 4,835 N.m) yields OD = 137 mm, t = 1.10 mm, Mass = 1.980 kg.

Independent Verification & Compliance Summary

- **100% Automated Test Suite Passing:** All 16 unit tests in tests/test_analysis.py pass in 0.02s, validating flight load derivations, known annular formulas, and trade studies.
- **Independent Closed-Form Arithmetic:** Full N/mm/MPa recalculation reproduces 14 finalist designs and 35 load scenarios with relative moment error < 1.56e-11.
- **Dense Circumferential Scan:** Co-located multi-axial stresses independently verified across 7,200 angular increments around cylinder perimeter.
- **Academic Limitations Honestly Declared:** NASA axial knockdown extrapolated for L/r = 25.4 > 5; end ring fitting mass separated from shell mass boundary.
- **Submission Deliverables Ready:** 1. Final Design Report (PDF) | 2. Python Code & Tests | 3. 2D CAD Engineering Drawing | 4. Presentation Slides.